

IMRAN NAWAR

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EDUCATION

Bachelor of Science in Computer Science

Islamia College Peshawar, Peshawar (Pakistan)

Sept 2020 - July 2024

CGPA: 3.78/4.00

- Advisor: [Dr. Muhammad Sajjad](#), PI Digital Image Processing Lab
- Thesis: Deep Learning Assisted Autonomous Navigation over Economical Hardware

EXPERIENCE

Digital Image Processing (DIP) Lab

Research Assistant

July 2024 - Present

Peshawar, Pakistan

- Co-authored three research papers: one on a cybersecurity framework for the Internet of Energy, published in IEEE Network (*Impact Factor: 6.3, Q1*); the second on autonomous navigation, under review at IJVAS; and the third on consumer devices, currently ready for submission to IEEE TCE.
- Worked on the UKRI Research Proposal with [Dr. Maqsood Mahmud](#) of Ulster University.
- Act as Teaching Assistant for Data Visualization (DSC 635) [🔗](#), Digital Image Processing (COMP416), Information Retrieval (COMP423), Artificial Intelligence (COMP315) [🔗](#) and Advanced Machine Learning (CSC 731) course. [\[Evaluation Material\]](#)

Corvit Networks

Artificial Intelligence Instructor

Dec 2024 - Aug 2025

Islamabad, Pakistan

- Delivered the **Artificial Intelligence (Machine Learning & Deep Learning)** course at Corvit Networks, as part of the Prime Minister's Special Initiative for Youth in collaboration with NAVTTC. [\[Dec 2024 - Mar 2025\]](#)
- Delivered the **HCCDA-AI** (Huawei Cloud Certified Developer Associate - Artificial Intelligence) course at Corvit Networks, as part of the Huawei's mission to empower youth across Pakistan in collaboration with NAVTTC. [\[May 2025 - Aug 2025\]](#)

Digital Image Processing (DIP) Lab

Undergraduate Student Researcher

Nov 2023 - June 2024

Peshawar, Pakistan

- Worked on autonomous navigation project for BS thesis.
- Designed and developed an autonomous navigation prototype on economical hardware (Jetson Nano).
- Developed an RFID-based door lock system using Arduino microcontroller.

PUBLICATIONS

[1] NeuroShield-IoE: A Cognitive, Quantum-Resilient, and Self-Evolving Cybersecurity Framework for the Internet of Energy (Paper)

Shehzad Ali*, [Imran Nawar](#)*, Ik Hyun Lee, Muhammad Sajjad, Aziz Alotaibi, Victor Hugo C. de Albuquerque, Khan Muhammad

Oct 2025

[IEEE Network](#)

[2] Democratizing Autonomous Driving Research via Economical Hardware (Poster) (Code)

[Imran Nawar](#), Wajahat Ullah, Muhammad Sajjad

June 2026

[Under Review at IJVAS](#)

[3] Privacy-Preserving Semi-Supervised Acoustic Anomaly Detection on Consumer Devices (Code)

[Imran Nawar](#), Shehzad Ali, Muhammad Sajjad, Khan Muhammad

[In Submission to IEEE TCE](#)

HONOURS AND AWARDS

- Recipient of a **Project Based Stipend Scholarship**, funded from an Australia's Economic Accelerator Innovate Grant, to undertake a Master of Applied Research at Victoria University, Melbourne, Australia
- Achieved **4.00/4.00 GPA** in Final Semester (Spring 2024) 2024
- Recipient of **FYP Research Grant** from Ignite NGIRI (PKR 88,644 grant) 2024
- Recipient of **Prime Minister's Laptop** under the PM's Youth Laptop Scheme (Phase III) 2023

PROJECTS

- [1] Talaash

Mar 2026 - July 2026

Tools: React, Vite, FastAPI, Python, ChromaDB

• Build a platform that helps researchers in finding research groups that truly match their interests and goals.
- [2] DIPCar: Autonomous Navigation over Economical Hardware

Mar 2024 - Oct 2024

Tools: Jetson Nano, SSD MobileNet, FCN ResNet34, TensorRT

• Designed and developed an autonomous navigation prototype on economical hardware (Jetson Nano).
- [3] Facial Emotion Recognition using CLIP encoder

Aug 2024 - Oct 2024

Tools: CLIP model, Streamlit, PyTorch

• Implemented and evaluated deep learning models for facial emotion recognition using the FER-2013 dataset, achieving real-time performance with fine-tuned *openai/clip-vit-base-patch32*.
- [4] Youtube Video Transcript Summarizer

June 2024

Tools: gemini-api, streamlit, youtube-transcript-api, python

• Developed a Streamlit web application that extracts YouTube video transcripts and generates concise, detailed notes using Google's Gemini API.

TECHNICAL SKILLS

Solved 150+ Leetcode problems

Languages: Python, C/C++, HTML/CSS

Frameworks & Libraries: PyTorch, Scikit-learn, Hugging Face, Ultralytics, TensorRT, Numpy, Pandas, Matplotlib, Seaborn, OpenCV, Streamlit, Jetson-Inference, fastAPI, ModelArts

Tools: VS Code, Curor, Antigravity, Git, GitHub, Jupyter, Kaggle, Google Colab, Vercel, Netlify, Linux, Roboflow, Draw.io, MS Office, Markdown, LaTeX

Core Skills: Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Neural Networks, Data Visualization, Data Structures & Algorithms, Technical Writing, Presentation Skills

VOLUNTEER EXPERIENCE

- **Financial Coordinator**, DIP Lab

Aug 2024 - Present

◦ Managing lab finances, including budgeting and expense tracking.

CERTIFICATIONS

- Computer Vision Basics

The State University of New York | Coursera - June 2025
- Neural Networks and Deep Learning

IBM | Coursera - Mar 2025
- Mathematics for Machine Learning: Linear Algebra

Imperial College London | Coursera - Jan 2025
- Getting Started with AI on Jetson Nano

NVIDIA - Sep 2024
- Machine Learning Specialization

DeepLearning.ai | Coursera - April 2024
- Fundamentals of Digital Image and Video Processing

Northwestern University | Coursera - Feb 2024
- Matrix Algebra for Engineers

The Hong Kong University of Science and Technology | Coursera - Dec 2023
- Python for Data Science, AI & Development

IBM | Coursera - Nov 2023
- Meta Front-End Developer Specialization

Meta | Coursera - Jul 2023
- Introduction to Python

Datacamp - Oct 2022

REFERENCES

Referee's are available on request